

Autonomous Fire Detection & Suppression System

Group 5

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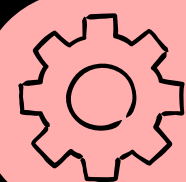
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The Problem: Outdated Equipment

Static sprinklers use universal agents — often making fires worse
Conventional sensors only alert; they can't act without human intervention

Classification Gap



✗ Water on Oil

Steam explosion scatters burning liquid, making it more hazardous

✗ Water on Electrical

Causes electrocution & new short circuits which only escalates the situation

✗ Water on Metal Fire

Strips oxygen from water — feeds the blaze.

Automation Gap

- **A fire doubles every minute**
- A system that is intelligent enough to *warn*, but too passive to *act*, allows a minor spark to become a lethal hazard.

**Problem
Analysis**

**Scale of the
Crises**

**Hardware
Components**

**Existing
Solutions**

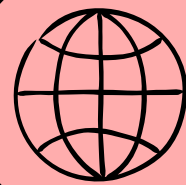
**Our
Solution**

**Cost & Market
Analysis**

✓ Fire Typologies and Targeted Extinguishing Agents

Types of Fire

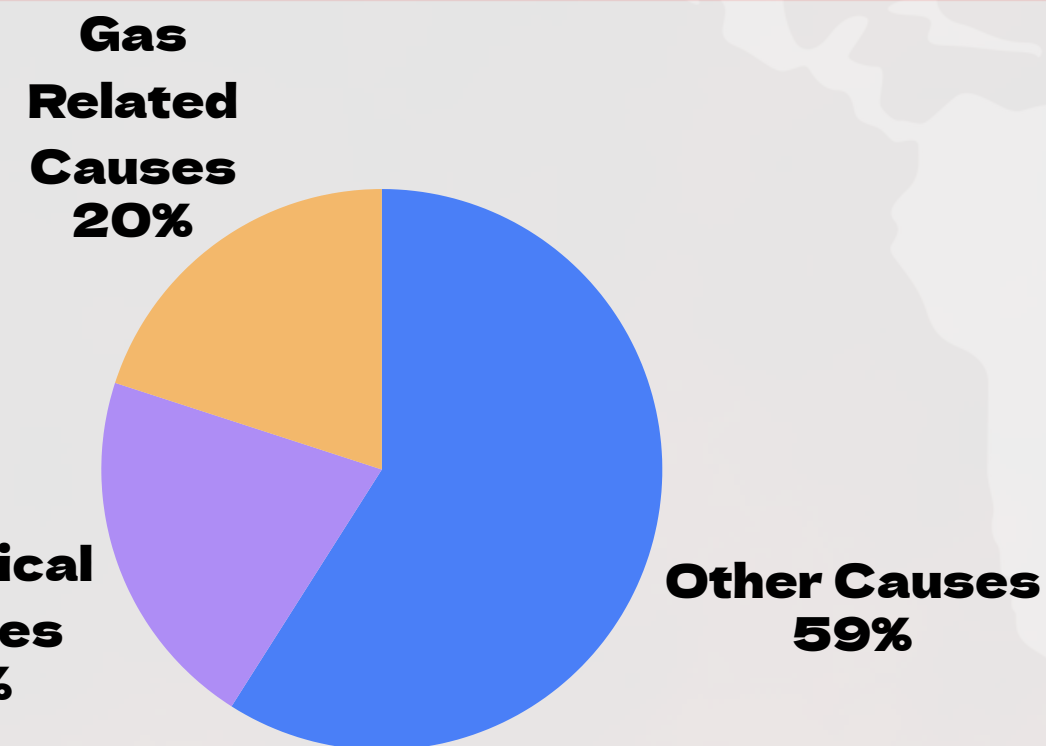
				
Class A	Class B	Class C	Class D	Class K/F
<u>Fuel</u> Wood, Paper, Cloth, Plastic, etc	<u>Fuel</u> Gasoline, Oil, Propane, Butane, etc	<u>Fuel</u> Electrical Equipment, Wiring, etc	<u>Fuel</u> Magnesium, Lithium, Sodium, etc	<u>Fuel</u> Animal and Vegetable fats used in cooking
<u>Extinguisher</u> Water or ABC Dry Chemical	<u>Extinguisher</u> CO2 or Foam or other Dry Chemical Agents	<u>Extinguisher</u> CO-2 or Specialized Dry Chemicals	<u>Extinguisher</u> Sodium Chloride based Powders	<u>Extinguisher</u> Wet Chemicals having Potassium Acetate



The Scale of the Crisis

In 2024, USA faced loss of around 20 billion USD, with a death toll of 2000, due to fire hazards.

fires that can not be extinguished with water



According to NFPA data, in USA, most accidents occur in home environment where it is easy to misclassify fire or respond late.

NEWS

UPDATE

TRP Game Zone Fire, Rajkot, India (May 2024):

A devastating fire erupted in a temporary children's play zone, claiming the lives of at least 27 people - including a newly wed couple and four children under 12.

Problem Analysis

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Cost & Market Analysis



Existing Solutions

Ceasefire Mini - Micro Environment Fire Suppression

Releases potassium aerosol to break the fire's chemical chain reaction without depleting oxygen.

Limitation

Uses a fixed agent and cannot automatically classify fire types.

ANSUL R-102 Restaurant Fire Suppression System

Heat triggers the discharge of ANSULEX to cool surfaces and block grease re-ignition.

Limitations

Specific to kitchen hazards; no dynamic gas or fire type analysis.

FirePro Condensed Aerosol Generator

Uses heat or electricity to deploy potassium aerosol, chemically halting combustion.

Limitations

Fixed aerosol agent; no fire type classification.

Flame Detection Based Fire Suppression Systems

IR/UV sensors detect flames to trigger the release of pre-selected agents (like CO₂).

Limitations

Detects flames only (not gas composition); suppression agent is fixed/non-adaptive.



**Ceasefire
System**



**ANSUL
System**

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Our Solution : SafeSense

The system utilizes an ESP32 microcontroller to analyze data from a sensor fusion suite, where an IR sensor detects flames and MQ-series sensors classify fire types (Class A or B) based on gas profiles. By processing these signals internally, the ESP32 eliminates latency, allowing for an immediate transition from hazard detection to targeted suppression.

Once the fire is identified, the ESP32 drives a high-torque pan-tilt turret using PWM signals to "peak hunt" and lock onto the fire's thermal core. It then triggers specific relays and solenoid valves to deploy the appropriate extinguishing agent. The architecture features centralized power management and voltage scaling to ensure high-precision operation and component safety.



**Problem
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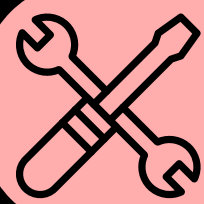
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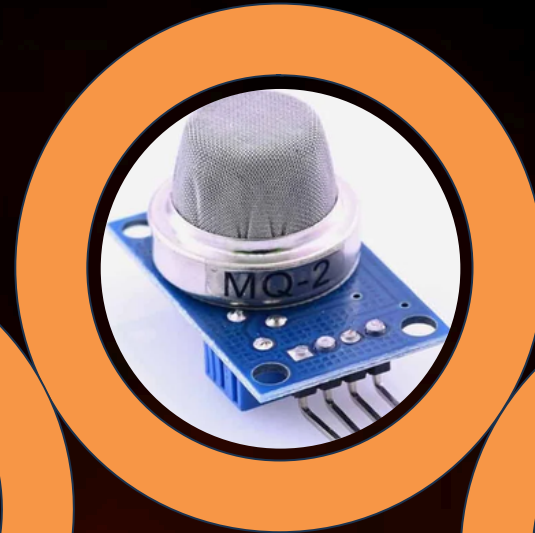


Hardware Components

1 MQ-2 Gas Sensor

Detects LPG, methane, hydrogen & smoke (200–10,000 ppm) → flags Class B fires

1



2 IR Flame Sensor

Confirms active flame at 760–1100 nm wavelength; prevents false triggers from gas leaks alone

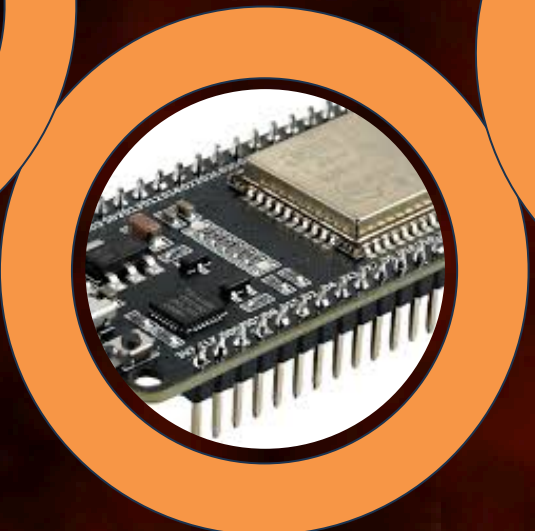
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3 Pan-Tilt Servo Turret

MG996R servos sweep and lock onto heat source via IR peak-hunting algorithm

3



4

MQ-135 Gas Sensor

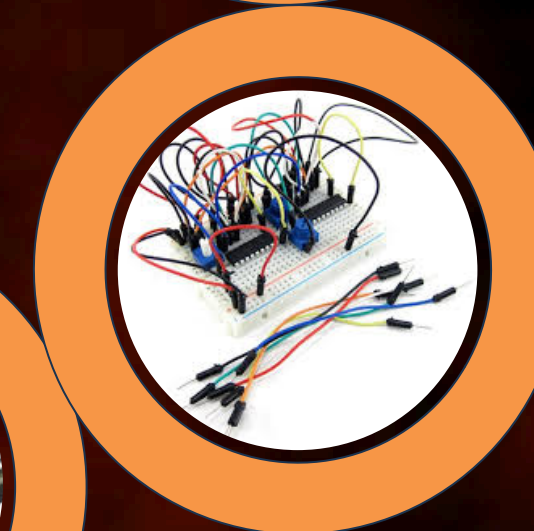
Detects CO₂, benzene & combustion byproducts → flags Class A solid-fuel fires



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ESP32

ESP32 classifies fire and handles real-time PWM motor control



6

Breadboard and Jumper Wires

They are used to connect the wiring of the the prototype.

Problem
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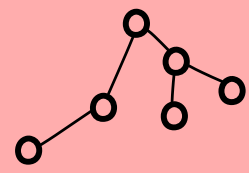
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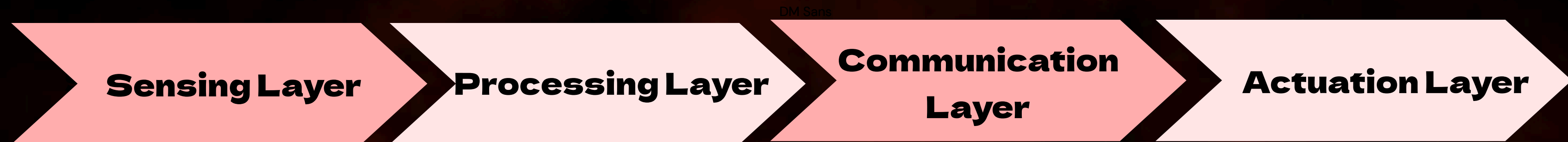
Cost & Market
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System Architecture & Data Flow

- **Sense:** Sensor fusion detects the fire
- **Think:** ESP32 classifies the fire type
- **Act:** ESP32 handles real-time targeting and suppression

Architecture:



Data Flow:



**Problem
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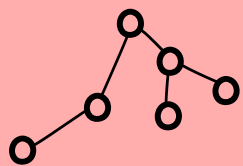
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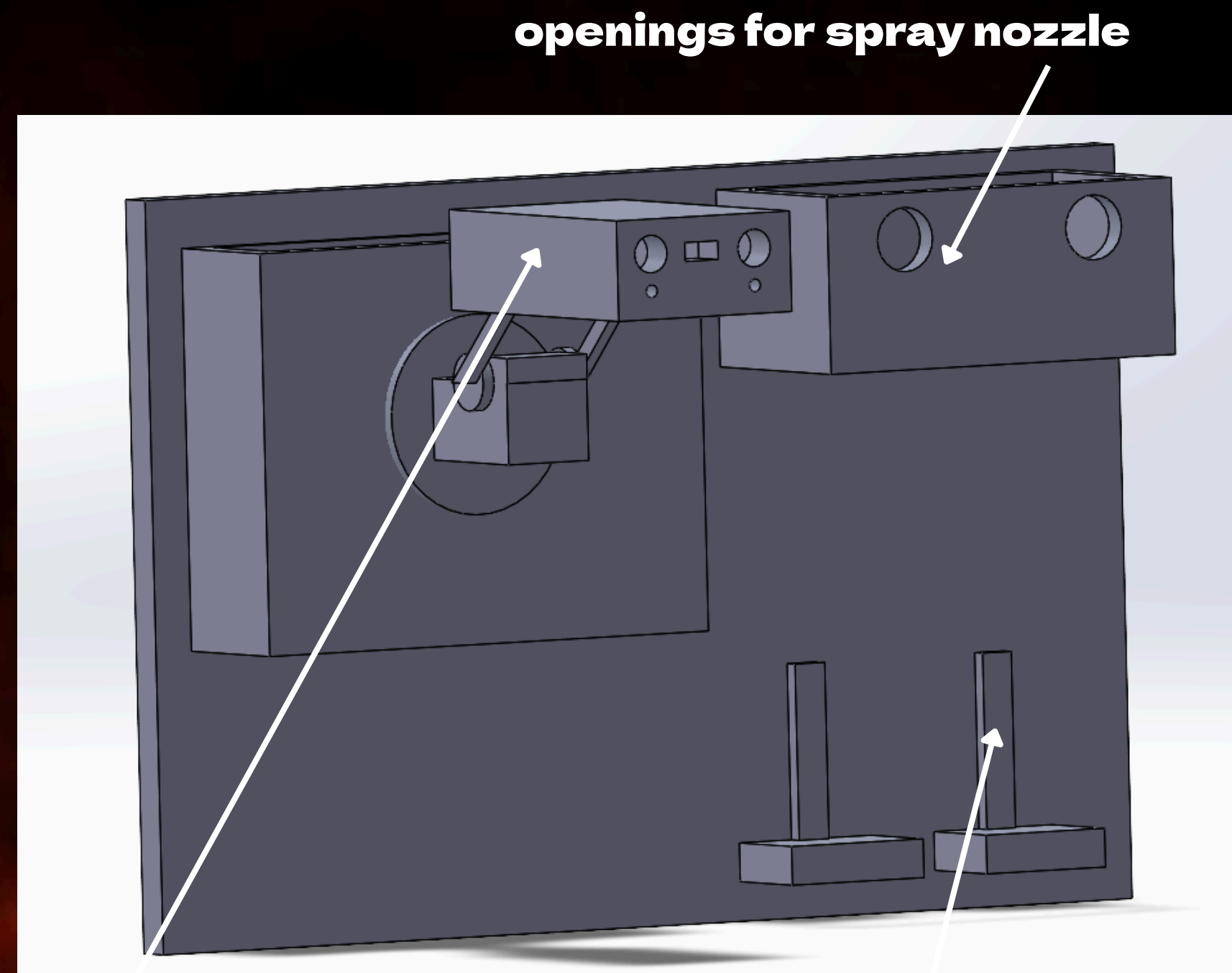
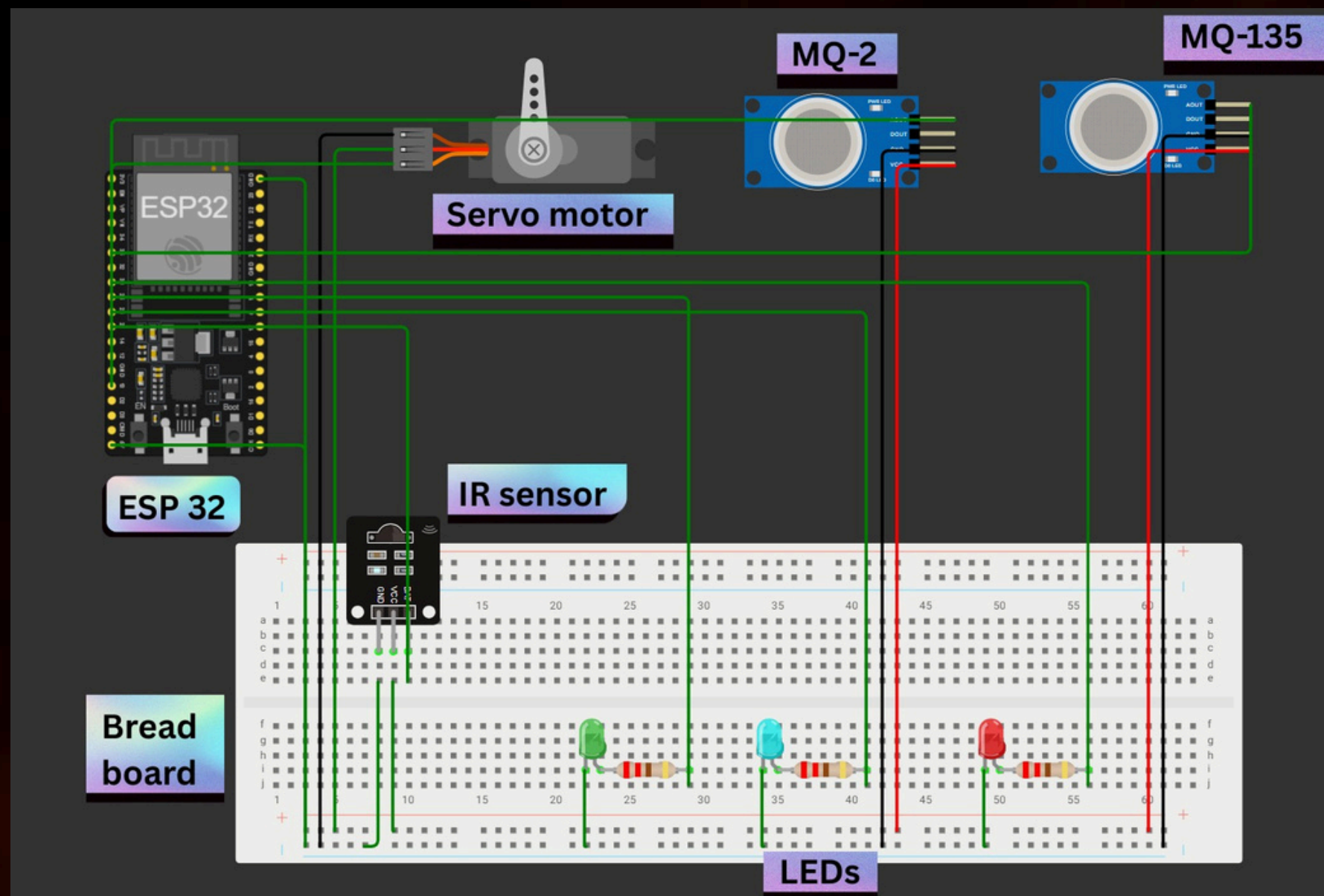
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Circuit Diagram & CAD Model



Adjustable structure for
sensor alignment

Extinguisher
Spray Bottle stand

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Cost & Market Opportunity

Prototype Cost

MQ Sensors, IR Sensor	₹400
ESP32	₹450
Servos (*2)	₹800
Cylinder	₹650
Misc. Components	₹300
Total	₹2600

Market Opportunities

Market Scale: The \$60.9 billion global fire safety market is projected to grow by 50% in eight years, alongside India's accelerating \$2.0 billion market.

To achieve industrial certification, standard sensors and microcontrollers must be replaced with Aspirating Smoke Detectors (VESDA), Multi-Spectrum IR/UV sensors, and PLCs for high-stakes reliability. Additionally, lightweight servos and canisters should be swapped for industrial BLDC drives and manifold-fed delivery systems housed in explosion-proof stainless steel.

USE CASES



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THANK YOU!